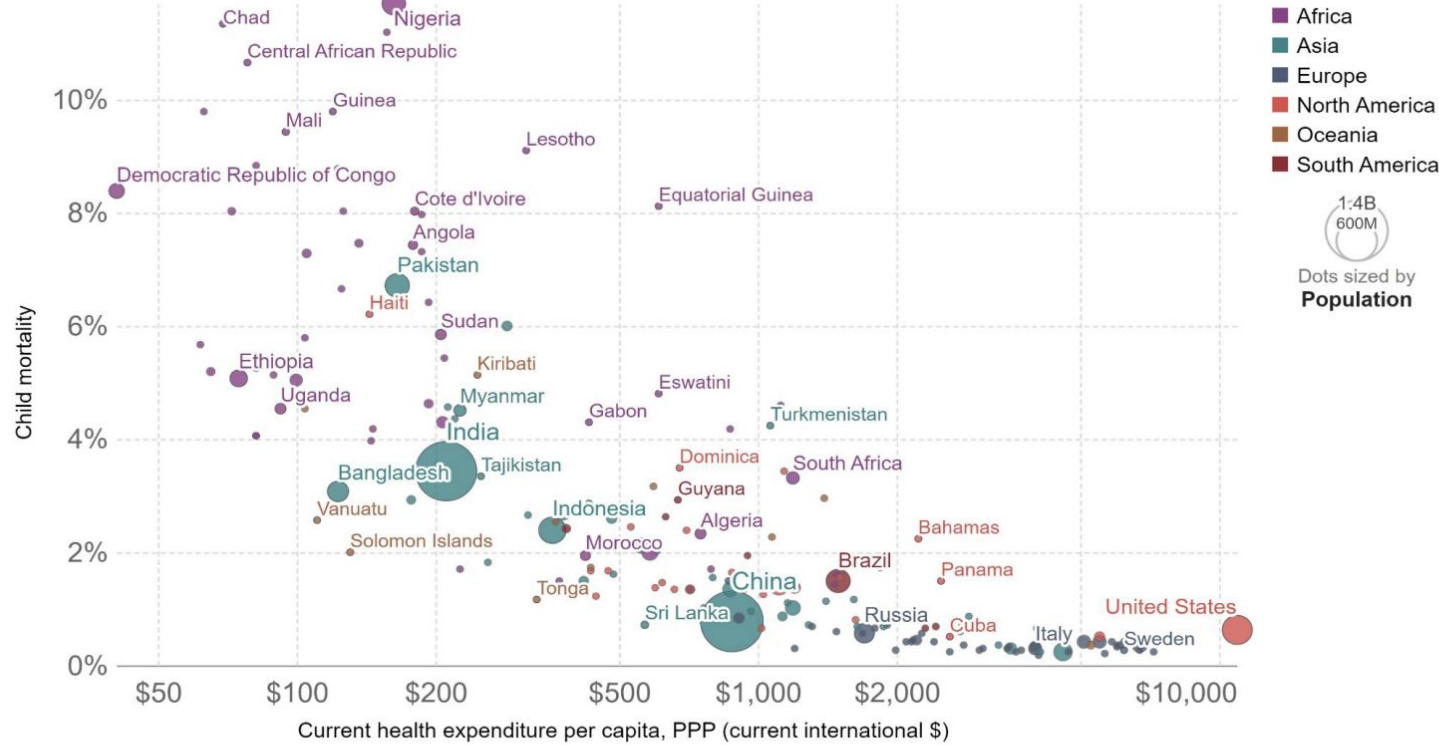


## 2.4 Population Dynamics

### Child mortality vs per capita health expenditure, 2019

Healthcare expenditure per capita is measured in current international-\$, which adjusts for price differences between countries. Child mortality is the share of newborns who die before reaching the age of five.

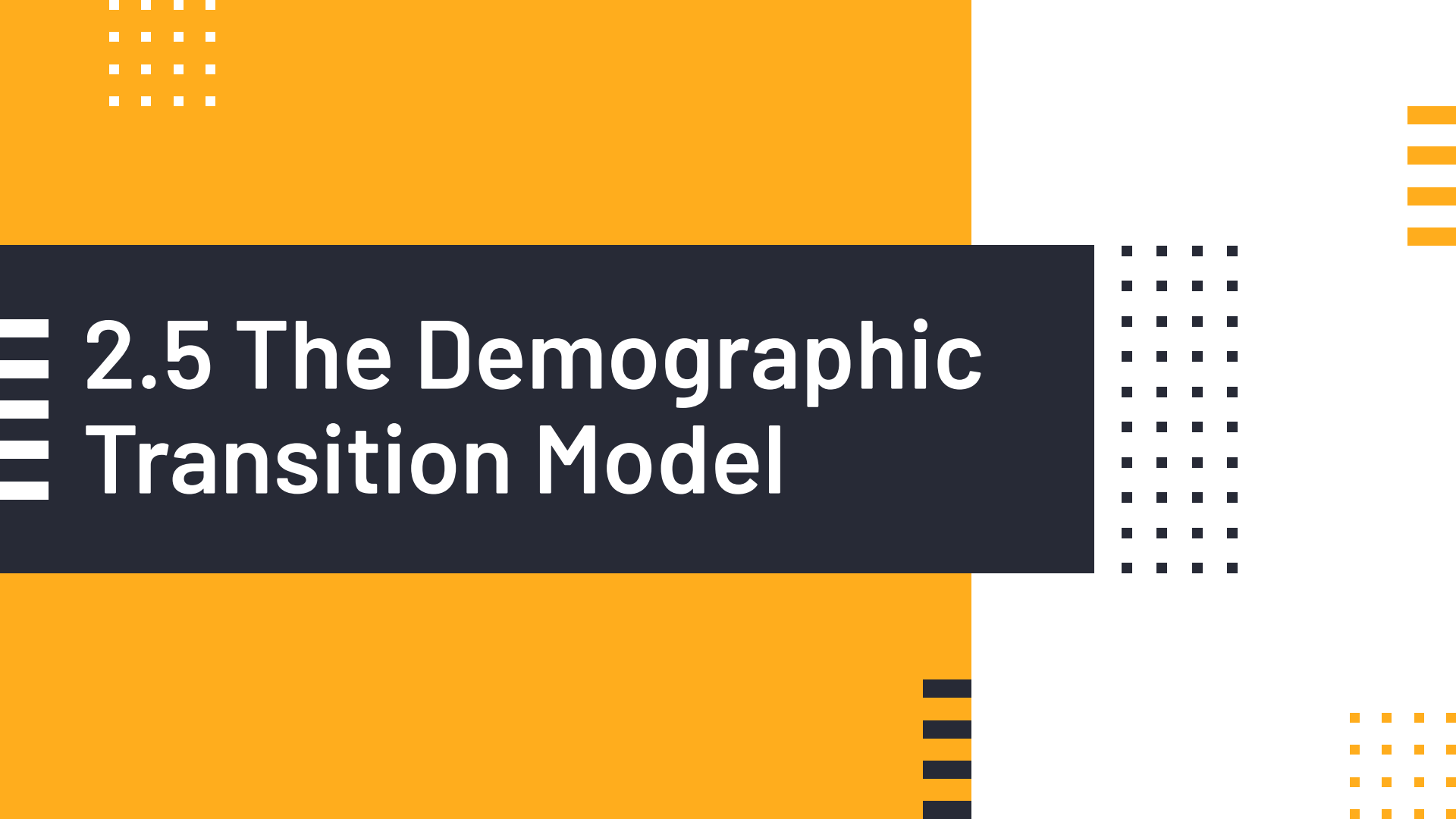


A. Describe an economic factor that influences child mortality rates.

B. Explain how child mortality rates in Africa affect life expectancy.

Source: UN Inter-agency Group for Child Mortality Estimation (via World Bank), World Health Organization (via World Bank)  
OurWorldInData.org/child-mortality/ • CC BY

source: [Our World in Data](https://ourworldindata.org)



## 2.5 The Demographic Transition Model

# Objective and Essential Learning

## 2.5 Explain theories of population growth & decline.

- The demographic transition model can be used to explain population change over time.
- The epidemiological transition explains causes of changing death rates.



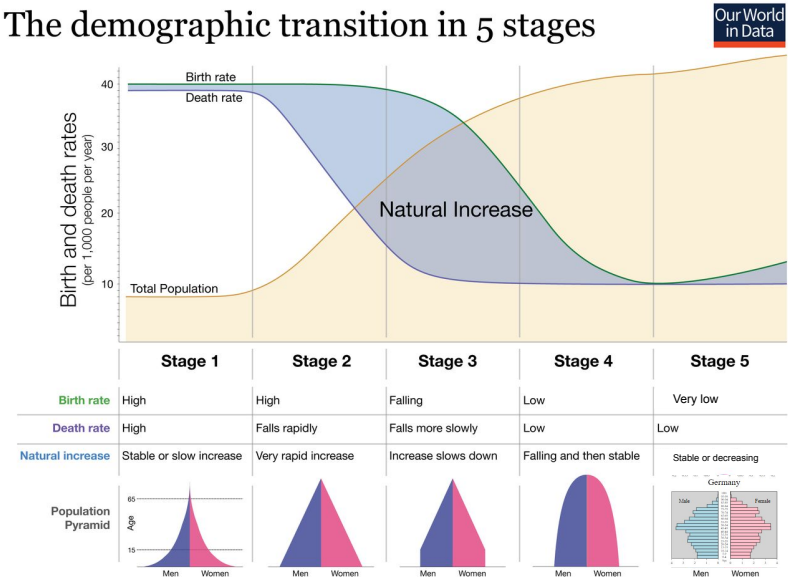
## 2.8 Explain how the changing role of females has demographic consequences in different parts of the world.

- Changing social values and access to education, employment, health care, and contraception have reduced fertility rates in most parts of the world.
- Changing social, economic, and political roles for females have influenced patterns of fertility, mortality...

# The Demographic Transition Model

- **Demography:** The study of population statistics.
- **Demographic Transition Model (DTM):** Model used by geographers to analyze and predict trends in population growth and decline including patterns of births, deaths and natural increase rates.

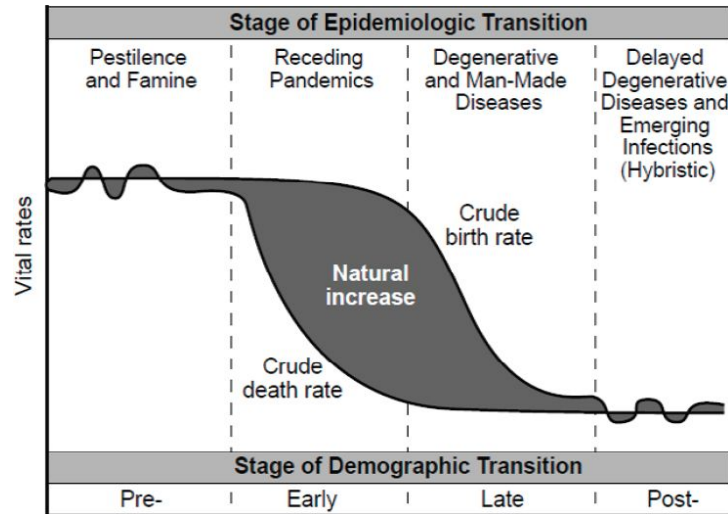
The demographic transition in 5 stages



The author Max Roser licensed this visualisation under a CC BY-SA license. You find more information at the source: <http://www.OurWorldinData.org/world-population-growth>

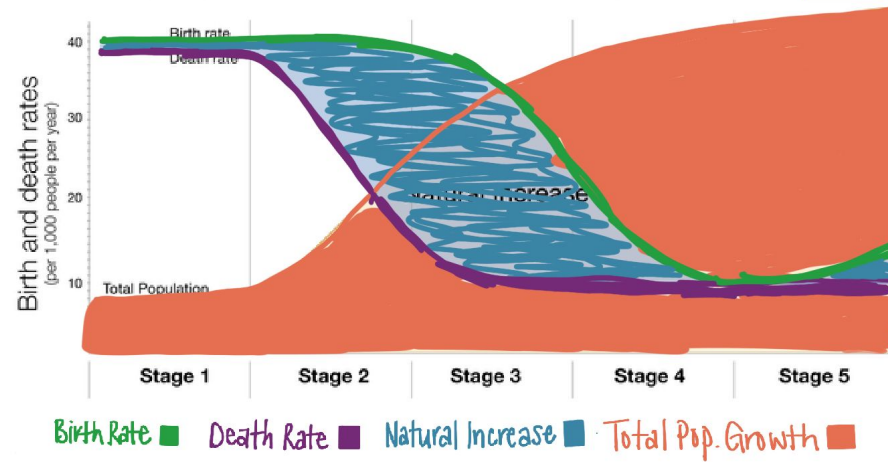
# The Demographic Transition Model

- **Epidemiological Transition Model (ETM):** Predictable stages in disease and life expectancy that countries experience as they develop which corresponds with the stages of the DTM.



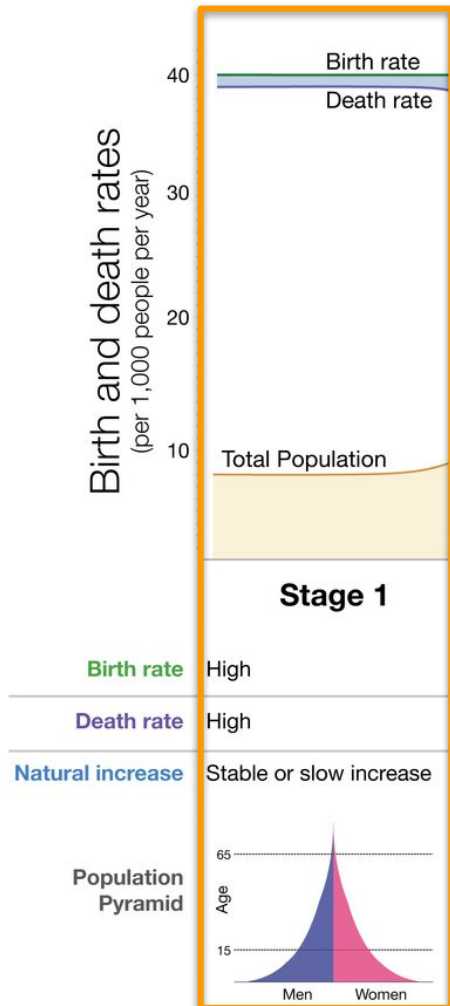
## The demographic transition in 5 stages

Our World  
in Data



Take a moment to annotate the essential aspects of the DTM model with **different colors**. You should annotate:

- Birth Rate
- Death Rate
- Natural Increase
- Total Population



## Characteristics of Stage One

### Reasons for CBR

- High TFR because children are not likely to survive long-term -> High IMR & CMR.
- Need for child labor in agricultural societies.

### Reasons for CDR

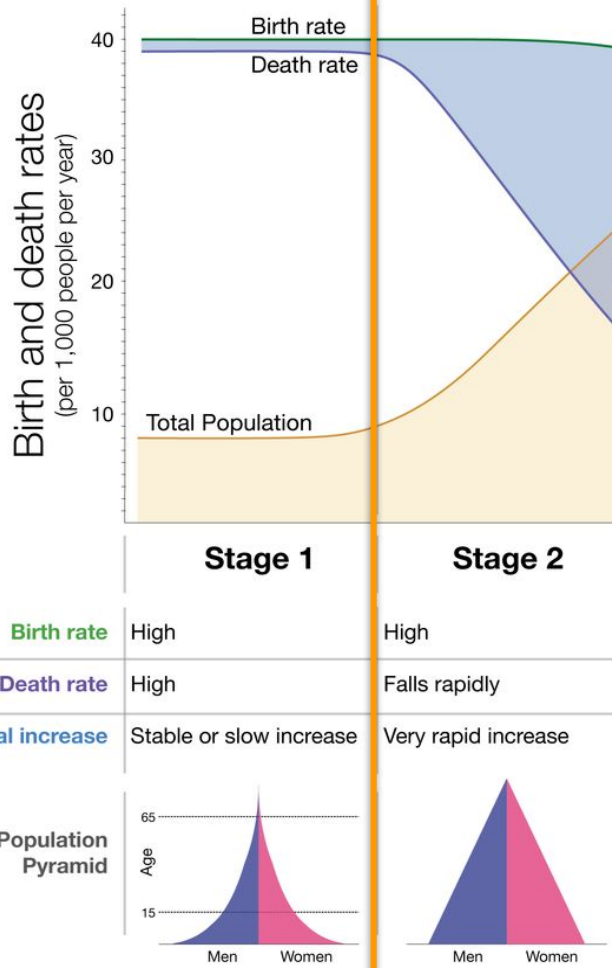
- Short life expectancy
- Famine, poor diet, inadequate housing, disease, warfare

### Natural Increase Rate

- Low growth

**Examples:** No countries are in this stage. Historic stage.

**Population Composition:** Youth dependency



## Characteristics of Stage Two

### Reasons for CBR

- Remains the same as previous stage

### Reasons for CDR

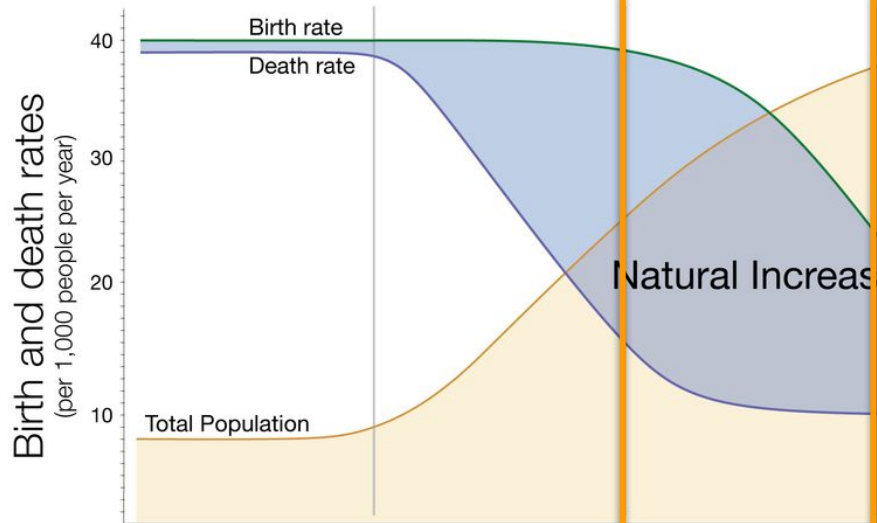
- Still a low life expectancy, but...
- Improved nutrition, sanitation and medicines.

**Natural Increase Rate:** Rapid growth

**Examples:** LDCs, Niger, Mali, South Sudan

**Population Composition:** Youth dependency





|                           | Stage 1                 | Stage 2             | Stage 3             |
|---------------------------|-------------------------|---------------------|---------------------|
| <b>Birth rate</b>         | High                    | High                | Falling             |
| <b>Death rate</b>         | High                    | Falls rapidly       | Falls more slowly   |
| <b>Natural increase</b>   | Stable or slow increase | Very rapid increase | Increase slows down |
| <b>Population Pyramid</b> |                         |                     |                     |

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## Characteristics of Stage Three

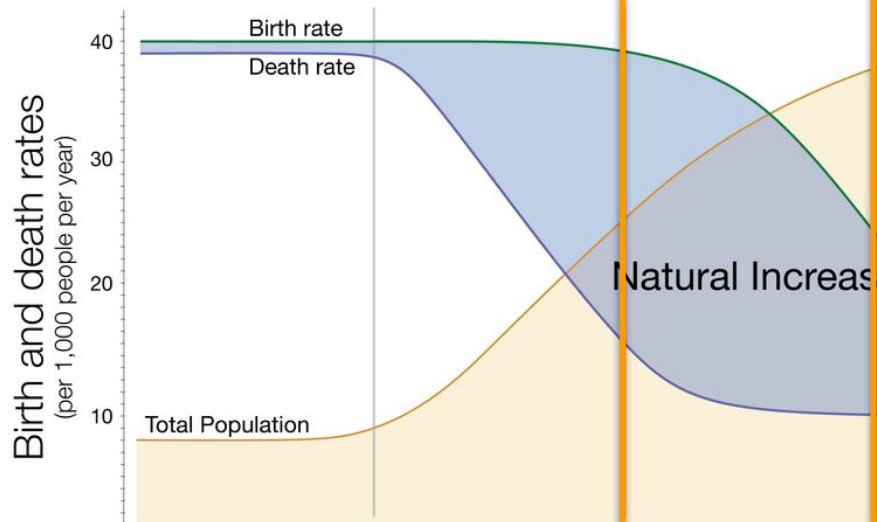
### Reasons for CBR

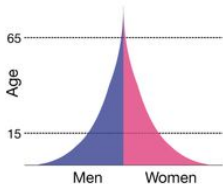
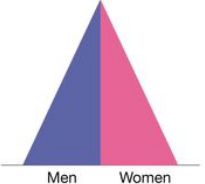
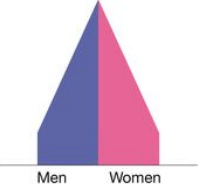
- Urbanization reduces the need for child labor - children \$\$, smaller living spaces.
- Increase in healthcare, female employment and education

### Reasons for CDR

- Life expectancy increases due to the progression of advancements from previous stage.

**Natural Increase Rate:** Growth slows



|                    | Stage 1                                                                            | Stage 2                                                                            | Stage 3                                                                             |
|--------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Birth rate         | High                                                                               | High                                                                               | Falling                                                                             |
| Death rate         | High                                                                               | Falls rapidly                                                                      | Falls more slowly                                                                   |
| Natural increase   | Stable or slow increase                                                            | Very rapid increase                                                                | Increase slows down                                                                 |
| Population Pyramid |  |  |  |

## Characteristics of Stage Three

**Examples:** Many Semi-Periphery countries like Turkey, Mexico, Indonesia, India

**Population Composition:** Society is still young, but there are more people surviving longer.

## Characteristics of Stage Four

### Reasons for CBR

- Same as previous stage.

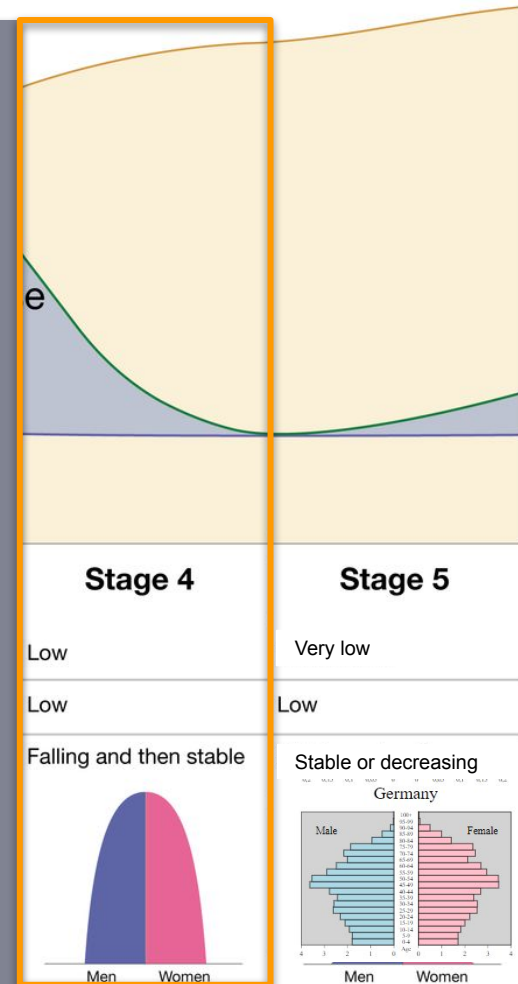
### Reasons for CDR

- Life expectancy continues to increase.

**Natural Increase Rate:** Falls and then stabilizes at low growth.

**Examples:** Majority of MDCs- China, Australia, Canada, South Korea, US

**Population Composition:** Beginning to shift to elderly dependency.



## Characteristics of Stage Five

### Reasons for CBR

- Replacement level (TFR: 2.1) OR
- Zero Population Growth (ZPG)

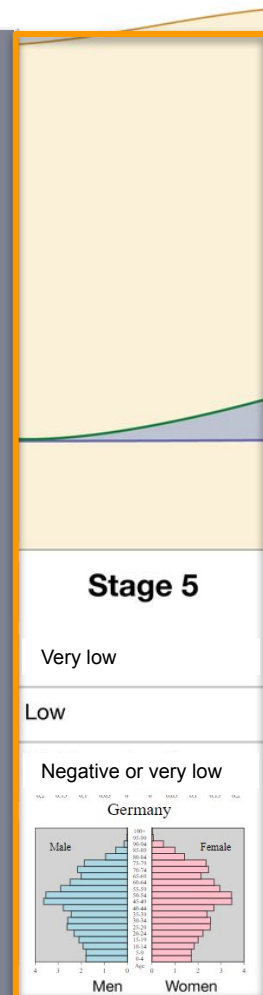
### Reasons for CDR

- Most medically advanced

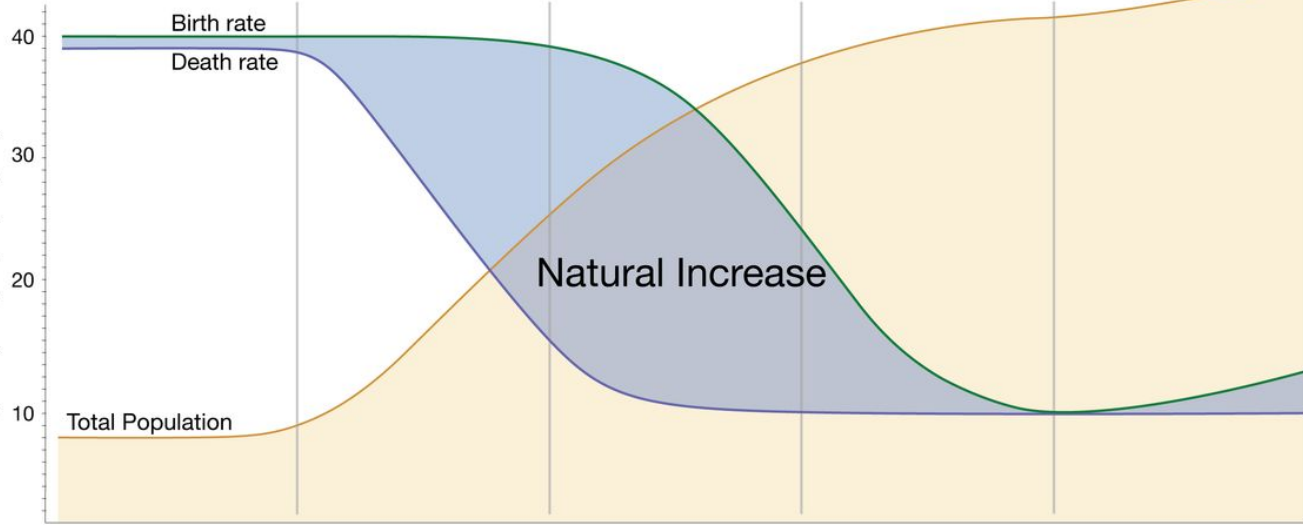
**Natural Increase Rate:** Negative or very low.

**Examples:** Japan, Germany

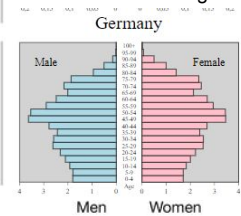
**Population Composition:** Most likely to have a high elderly dependency ratio.



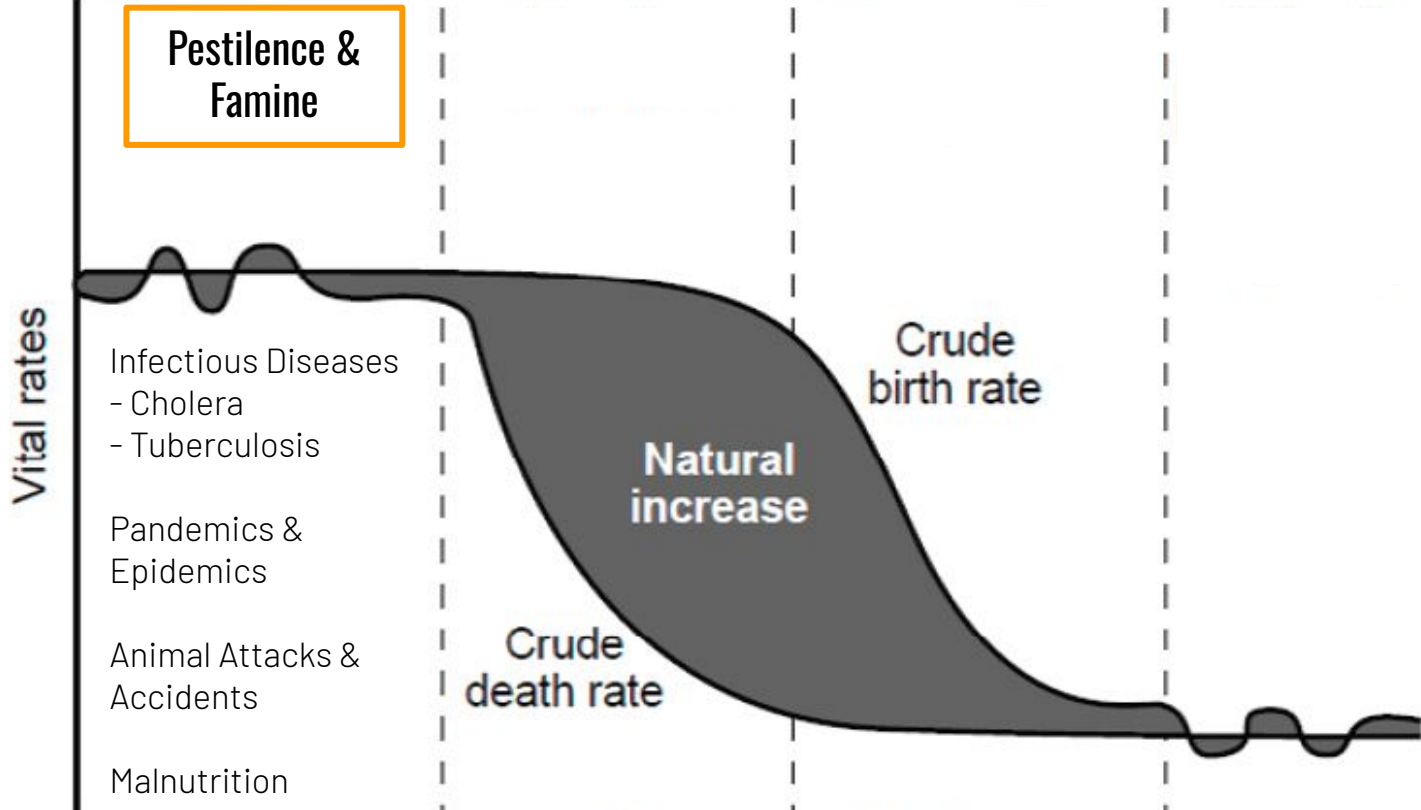
Birth and death rates  
(per 1,000 people per year)



|                           | Stage 1                 | Stage 2             | Stage 3             | Stage 4                 | Stage 5              |
|---------------------------|-------------------------|---------------------|---------------------|-------------------------|----------------------|
| <b>Birth rate</b>         | High                    | High                | Falling             | Low                     | Very low             |
| <b>Death rate</b>         | High                    | Falls rapidly       | Falls more slowly   | Low                     | Low                  |
| <b>Natural increase</b>   | Stable or slow increase | Very rapid increase | Increase slows down | Falling and then stable | Stable or decreasing |
| <b>Population Pyramid</b> |                         |                     |                     |                         |                      |



## Stage of Epidemiologic Transition



## Stage of Epidemiologic Transition

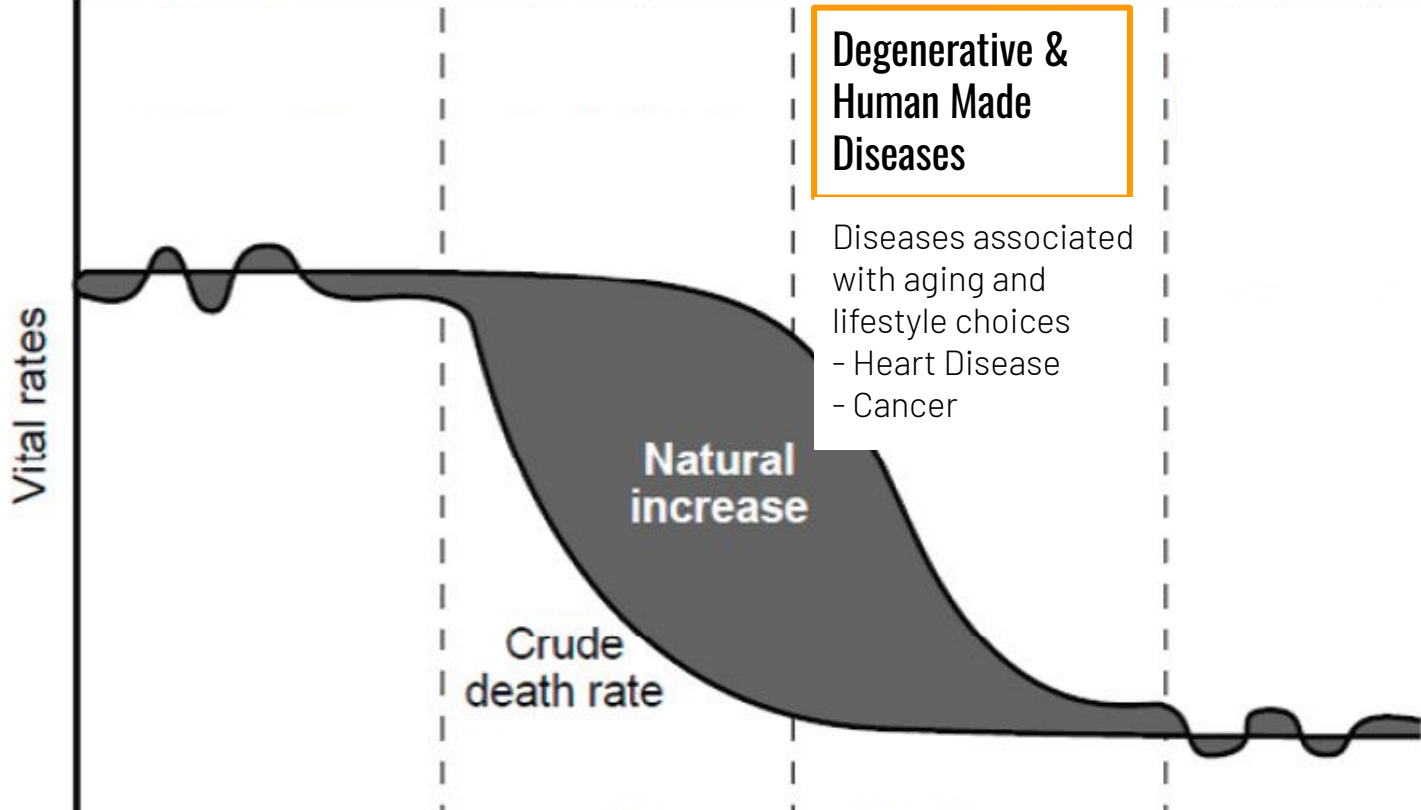
### Receding Pandemics

Vital rates

Pandemics and infectious diseases decline due to new medical advances, sanitation and improved nutrition.

Crude birth rate

## Stage of Epidemiologic Transition





## Stage of Epidemiologic Transition

Vital rates

Crude  
death rate

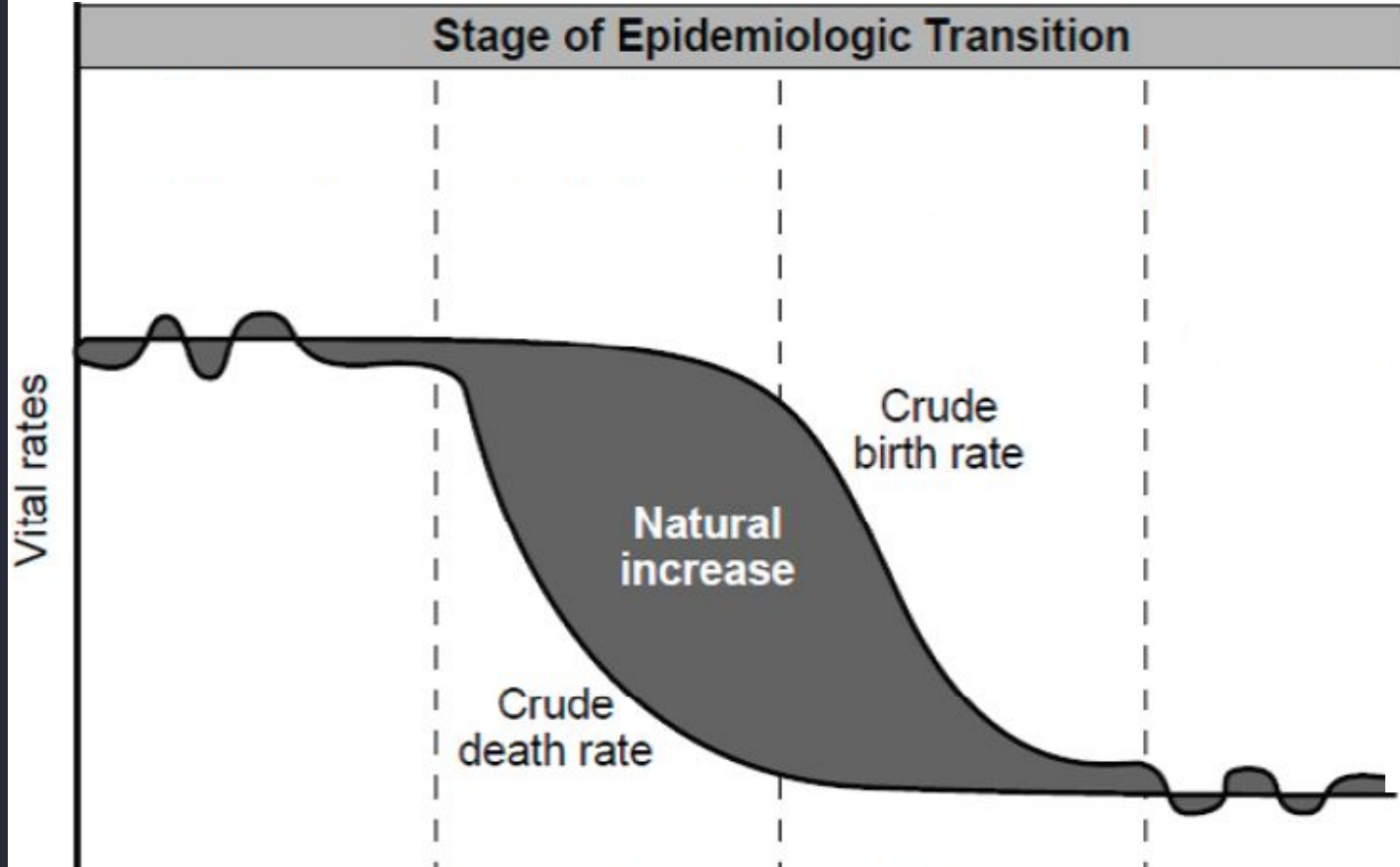
Natural  
increase

Crude  
birth rate

### Delayed Degenerative Diseases

Diseases associated with aging and lifestyle choices can be delayed with medical advancements.

Alzheimer's  
Dementia



## Reemergence of Infectious Diseases

Bacteria and parasites become resistant to antibiotics and vaccines.

Reemergence of infectious diseases.

Ebola  
COVID-19

# Objective and Essential Learning

## 2.5 Explain theories of population growth & decline.

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- The epidemiological transition explains causes of changing death rates.

